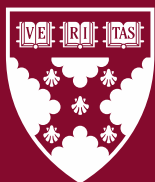


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CYBORGS, CENTAURS AND SELF-AUTOMATORS: THE THREE MODES OF HUMAN-GENAI KNOWLEDGE WORK AND THEIR IMPLICATIONS FOR SKILLING AND THE FUTURE OF EXPERTISE

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ABSTRACT

Research on human–AI collaboration in professional settings has focused primarily on isolated interactions between knowledge professionals and AI, usually at the point of decision making. Yet generative AI broadens these interactions into an ongoing conversational process that unfolds across the full workflow. In a field study of 244 global management consultants from the Boston Consulting Group, we investigated the human–GenAI decision-making workflow and found that knowledge professionals engaged in three distinct modes of human–AI knowledge co-creation: Fused Knowledge Co-Creation, Directed Knowledge Co-Creation, and Abdicated Knowledge Co-Creation. These modes correspond to three types that emerged in practice: “Cyborgs,” “Centaur,” and “Self-Automators.” Across these modes, we identify two fundamental questions that structure any collaborative problem-solving dynamic between human and machine: Who selects what needs to be done? and Who identifies how it gets done? These questions reveal the underlying architecture of human–AI collaboration, showing not only *what* GenAI is used for but also *who* steers the workflow and how the division of labor unfolds in practice. As a result of these emergent modes of working with GenAI, Centaurs were upskilling themselves and increasing their current domain expertise. Cyborgs were newskilling themselves in GenAI-related capabilities, acquiring a new area of AI expertise. Self-Automators were not increasing either their domain expertise or AI expertise. We show how what seems to be the same “human-in-the-loop” workflow for decision making is actually enacted in three ways with significantly different implications for skilling and the future of work.*

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INTRODUCTION

Managers and professionals across industries are now confronting an urgent question: whether, and how, to use generative AI (GenAI) in their daily problem-solving. As the technology spreads rapidly through organizations, debates have intensified not only over its potential to transform productivity and creativity, but also over what it will mean for the future of professional expertise and organizational capability. For many firms, the issue is no longer one of simple adoption but of reimagination: how should work itself be reorganized when a machine can participate in the problem-solving process?

At the center of this transformation lies a critical tension between automation and augmentation. Will professionals use GenAI to amplify their cognitive and creative capabilities, or will they gradually cede core elements of judgment and problem definition to the machine? As organizations experiment with AI integration, they confront a paradox: technologies designed to extend human potential can, if over-delegated, erode the very skills and discernment that make expertise valuable.

Yet much of this adoption is happening informally and unevenly. Many workers use so-called “Shadow AI,” the unsanctioned and improvised use of AI tools by employees (Ross, Hibbert, & Moss, 2025; Catherine et al., 2025). These bottom-up practices suggest that AI’s diffusion is outpacing formal governance, creating new organizational risks even as it surfaces latent demand for augmentation and experimentation. Large-scale reports from OpenAI (Chatterji et al., 2025) and Anthropic (2025) underscore the magnitude of this transformation: ChatGPT alone has reached roughly 10 percent of the world’s adults, with usage concentrated among highly educated professionals, and Claude has spread most rapidly in science, education, and consulting. Taken together, these developments highlight a critical inflection point: the question is no longer whether GenAI will enter knowledge work, but how it will reshape the conditions of professional problem-solving itself.

LLMs and GenAI change the dynamics of human–AI collaboration in potentially new and unexplored ways. Most studies of human–AI collaboration have focused on predictive AI systems that provide recommendations, highlighting barriers such as threats to autonomy and frustrations with opacity (e.g., Anthony, 2021; Christin, 2017; Pachidi et al., 2021; Valentine & Hinds, 2022; Van den Broek et al.,

2021; Waardenburg, Huysman, & Sergeeva, 2022). Such work has illuminated how professionals in domains as diverse as healthcare, criminal justice, finance, and policing struggle to integrate predictive AI into their decision making (e.g., Bechky, 2020, 2021; Karunakaran, 2024; Lebovitz, Levina & Lifshitz, 2021; Lebovitz, Lifshitz, & Levina, 2022; Waardenburg, Huysman & Agterberg, 2021, Waardenburg, Huysman, & Sergeeva, 2022). Building on this foundation, research has begun to examine how AI reshapes problem-solving in organizations, raising new questions about where expertise resides and how decisions are made (Faraj et al., 2018).

The nature of human–AI interaction is fundamentally different with LLMs and GenAI. Predictive AI interactions are usually task-specific, “narrow,” and tied to a single critical moment of problem-solving or decision-making recommendation. GenAI, by contrast, is conversational and iterative, allowing professionals to collaborate with it across the full arc of problem-solving workflow: from framing the problem, to analyzing data, to iterating, extending, evaluating, and reporting. This shift transforms not only what professionals do, but how they think and learn. GenAI enables professionals to have an ongoing conversational relationship with AI, in which the boundaries of human and machine work become negotiated across an entire workflow. Yet beyond changing what professionals do, GenAI may also be changing who they become, reshaping how they learn, develop expertise, and build new forms of skill.

Recent studies point to GenAI’s positive impact on performance and efficiency (Brynjolfsson, Li & Raymond, 2023; Csaszar, Ketkar & Kim, 2024; Dell’Acqua et al., 2023; 2025; Dillon et al., 2025; Noy & Zhang, 2023; Otis et al., 2023). Yet these studies largely capture short-term productivity and performance outcomes while we know little about the underlying knowledge-work-processes themselves. Our study opens this black box by examining the conversational fabric of human–AI collaboration and analyzing how professionals use GenAI across the full problem-solving workflow. Drawing on 244 consultant–AI problem-solving sessions at Boston Consulting Group (BCG), comprising 4,975 human–AI interactions and follow-up interviews, we reveal how professionals structure and lead collaborative

workflows with GenAI—deciding which tasks to delegate, which to retain, and how to orchestrate the overall problem-solving process.

We shed new light on this debate by examining the knowledge-work process itself: what professionals actually *do* when they use GenAI. Through exploration of the full problem-solving workflow, we reveal hidden patterns in how they structure, delegate, and in some instances, verify their collaboration with the system. What's striking is that although every consultant had access to the same tools and the same task, their choices about when to engage GenAI and how much authority to give it differed dramatically.

We find that professionals may engage in three distinct modes of knowledge co-creation processes – Fused Co-Creation, Directed Co-Creation, and Abdicated Co-Creation– that create three types of corresponding human-AI interactions: Cyborg, Centaur, and Self-Automator. Each mode and corresponding type has distinct implications for professionals' skilling, leading respectively to newskilling, upskilling, or no skilling. By diving deeply into the conversations themselves, we lay the ground to develop new theories on how humans interact with AI, and with what consequences for professional expertise and the future of work.

CURRENT LITERATURE ON HUMAN-AI COLLABORATION

Existing studies of the human–AI collaboration process have provided crucial insights into how professionals interact with AI systems at the point of recommendation. This body of work, which has focused primarily on algorithms (Anthony, 2021; Cameron, 2022; Cameron & Rahman, 2022; Rahman, 2021; Rahman & Valentine, 2021) and predictive AI (e.g., Anthony, 2021; Christin, 2017; Pachidi et al., 2021; Valentine & Hinds, 2022; Van den Broek et al., 2021; Waardenburg, Huysman, & Sergeeva, 2022), has advanced our understanding of how professionals navigate challenges such as threats to autonomy (e.g., Pachidi et al., 2021; Valentine & Hinds, 2022), algorithmic opacity (e.g., Lebovitz, Lifshitz, & Levina, 2022; Rahman, 2021), and the complexity and low usability of many algorithmic systems (Anthony, 2021; Mazmanian, 2013). We build directly on this foundation, extending it to examine how these dynamics unfold when collaboration with AI is no longer confined to discrete recommendation

moments but instead becomes conversational and continuous. In doing so, we treat prior research as a critical theoretical platform for understanding how GenAI transforms, not replaces, the forms of human–AI interaction that scholars have long examined. By shifting attention from point-based collaboration to full workflow engagement, our study contributes to this literature by theorizing the iterative, co-creative processes through which professionals and AI jointly construct knowledge.

Before describing our research setting, we first review current literature on human-AI collaboration with predictive AI, which details recommendation-specific barriers and processes associated with accomplishing human-AI collaboration. Next, we bring in insights from the literature on human-AI co-creation to explain three types of human-AI co-creation, what we call *Fused Co-Creation*, *Directed Co-Creation*, and *Abdicated Co-Creation* that are critical to understanding the future of problem-solving work in professional organizations.

Human-AI Collaboration: Overall and Barriers

As AI systems have been increasingly implemented in organizations, scholars have investigated dynamics associated with human-AI collaboration. They have highlighted the barriers to collaboration as well as the key processes that help to address these barriers in fields ranging from healthcare (Lebovitz, Levina & Lifshitz, 2021; Lebovitz, Lifshitz, & Levina, 2022; Pine & Mazmanian, 2017; Pine et al., 2022) to high technology (Rahman, 2021; Rahman & Valentine, 2021; Rahman, Karunakaran & Cameron, 2024; Valentine & Hinds, 2022; Van den Broek, Sergeeva & Huysman, 2020, 2021) to finance (Anthony, 2018, 2021; Beane & Anthony, 2024) to consumer products (Jia et al., 2024; Pachidi et al., 2021), to criminal justice (Bechky, 2020, 2021; Christin, 2017; Karunakaran, 2024; Waardenburg, Huysman & Agterberg, 2021; Waardenburg, Huysman, & Sergeeva, 2022). Three key barriers to human-AI collaboration have been shown to be professionals' experience of threat to autonomy, their frustration with AI opacity, and their concern that AI systems can be difficult to work with because of their complexity and low usability. First, AI systems can threaten professional autonomy because such systems create connections and find patterns between data points to arrive at insights that potentially rise above human knowledge (e.g., Kim et al., 2024; Pachidi et al., 2021; Valentine & Hinds, 2022; Van den Broek

et al., 2021). Such systems can lead knowledge workers to reject algorithmic recommendations, because they view algorithms as threatening their expert decision-making and autonomy (Christin, 2017; Lebovitz, Lifshitz, & Levina, 2022; Pachidi et al., 2021; Valentine & Hinds, 2022).

Second, AI systems can be opaque because many of them do not follow rule-based patterns; they uncover hidden relationships that are far too large and complex for humans to decipher (e.g., Burrell, 2016; Pasquale, 2015). For example, algorithms that highlight likely crime areas are being used by members in police organizations to make patrol decisions (Brayne, 2017; Karunakaran, 2024; Waardenburg, Huysman & Agterberg, 2021). However, this same opacity can present challenges related to concerns about users being able to evaluate the accuracy of the model (e.g., Anthony, 2021; Christin, 2020) and understanding model recommendations, particularly when the model generates unexpected or surprising outputs that contradict the intuition of decision makers (e.g., Lebovitz, Lifshitz, & Levina, 2022; Waardenburg, Huysman, & Sergeeva, 2022).

Third, AI systems can be difficult to work with due to their complexity and low usability, which can discourage professionals from incorporating them into their routines. Unlike many traditional tools, AI systems often lack visible interfaces, update dynamically, and operate through logic that even experienced users struggle to interpret (Mazmanian, 2013; Lebovitz, Lifshitz, & Levina, 2022). These qualities make it harder for professionals to build fluency with the system through experience or feedback (Bailey & Barley, 2011; Anthony, 2021). Adopting AI may also require shifts in how expertise is demonstrated and recognized, unsettling familiar ways of working and introducing ambiguity into established hierarchies (Kaplan et al., 2017; Pachidi et al., 2021). As a result, some professionals avoid or resist these tools, not only due to their design, but because using them can disrupt the social order of expertise in occupational life (Leonardi, 2007; Kellogg, 2009).

Human-AI Collaboration Processes

Scholars of human-AI collaboration in professional settings have detailed how professionals may address these barriers in four primary ways. First, technology developers and domain experts can collaborate around AI, or not, during the development process (Pachidi et al., 2021; Waardenburg,

Huysman & Sergeeva, 2022, Valentine & Hinds, Van den Broek et al., 2021). For example, Pachidi and colleagues (2020) demonstrate how technology developers may fail to engage meaningfully with domain experts during development, while Van den Broek and colleagues (2021) demonstrate how technology developers can engage in hybrid learning with domain experts and AI.

Second, domain experts can collaborate with one another around the use of algorithms (e.g., Anthony, 2021; Rahman & Valentine, 2021; Karunakaran, 2022). For example, Anthony (2021) details how junior and senior investment bankers productively engaged in understanding algorithmic outputs rather than black-boxing them. And Rahman & Valentine (2021) elaborate how client managers can engage in collaborative problem-solving with contractors around algorithmic recommendations using a process they call collaborative repair.

Third, domain experts can collaborate with AI by accepting or rejecting algorithmic recommendations around specific tasks. For example, Rahman (2021) explores how workers respond to algorithmic recommendations they cannot fully understand or influence. Cameron (2022) and Rahman and Cameron (2022) explain how Uber drivers can resist control by accepting or rejecting algorithmic recommendations to accept rides, and Rahman, Karunakaran & Cameron (2024) detail how high-skill labor market platform workers can resist algorithmic control by accepting or rejecting algorithmic recommendations to accept particular assignments.

Fourth, professionals can employ engaged AI interrogation (Lebovitz, Lifshitz, & Levina, 2022), a process that unfolds at the point of interaction with the AI's recommendations. Through this practice, experts actively question and interpret algorithmic outputs, comparing them to their own experiential knowledge and synthesizing the two forms of expertise to reduce uncertainty. Engaged AI interrogation thus enables professionals to critically engage with AI tools rather than passively accept their outputs, addressing challenges of opacity while helping preserve professional autonomy.

Thus, according to the prior literature, once an AI model is out of development and implemented in daily work, the process of human-AI collaboration occurs around discrete tasks, at the point of

recommendation, and entails a professional choosing to accept or reject the AI-informed recommendation.

Insights from the Human-AI Co-Creation Literature and the Upskilling Literature

In sum, the existing literature on human-AI collaboration elaborates how end user professionals engage with AI at the point of recommendation. Thus, this literature helps to explain, in our case, how AI provided professionals with narrow assistance around the constrained and specialized task of recommending to consultants which brand a CEO should focus on. We find, in addition, that there is merit to focusing on the conversational, iterative process across the consultants' full problem-solving workflow.

To better understand the new human-AI collaboration now emerging across organizations, we draw on insights from the literature on human-AI co-creation and employee upskilling. As GenAI tools rapidly permeate professional work, they are reshaping how ideas are produced, refined, and realized, often in ways that we do not yet fully understand. The human-AI co-creation literature offers a valuable lens for examining these changes, explaining how humans and GenAI can act as co-creative partners engaged in continuous, real-time improvisation that enriches the process of solution generation (e.g., Davis et al., 2015; Bown & Brown, 2018). In this view, professionals often begin with a loose intention, an initial framing or question that both constrains and suggests possible directions for exploration. GenAI collaborates with them on a shared creative product, combining human intuition and contextual understanding with the system's generative capacity. Rather than following a fixed plan, this collaboration unfolds experimentally, as the human and GenAI engage in a dynamic call-and-response exchange in which each builds on the other's prior contribution. Through this iterative process, professionals use GenAI across the full arc of problem-solving, from problem definition to analysis, iteration, and refinement, transforming both the process and pace of knowledge work.

As the literature on barriers to human-AI collaboration shows, AI can threaten professional autonomy and introduce uncertainty about expertise, leaving many professionals unsure of how to sustain agency in their work. While the human-AI co-creation literature offers valuable frameworks for

understanding how humans and intelligent systems can act as collaborative partners, it also underscores a core tension: we do not yet fully understand how these dynamics play out with GenAI. We also do not fully understand the skill-building dynamics associated with human-AI collaboration with GenAI.

While the literature on human-AI co-creation helps us to better understand the three types of human-AI co-creation we observed (Fused, Directed, and Abdicated Co-Creation), the literature on skill-building helps us to better understand the skills-related dynamics associated with these processes.

Upskilling refers to the process by which workers acquire new or enhanced skills that build upon their existing capabilities to meet evolving job demands, especially in response to technological change (e.g., Anthony, 2021; Beane, 2024; Beane & Anthony, 2024; Kaynak, 2019; 2024a; 2024b; Kellogg et al., 2021; Mazmanian & Beckman, 2018). In our study, professionals who engaged in Directed Co-Creation (Centaurs) upskilled themselves in their domain-related capabilities. Newskilling entails professionals developing entirely new capabilities that enable them to engage with emerging technologies and contribute to innovation (Kellogg, 2021). This could involve bringing on board employees with novel skills such as robotics skills in surgery (Beane, 2019), data science skills in high tech (Stice-Lusvardi, Hinds, & Valentine, 2024), or predictive analytics skills in finance (Anthony, 2018). In our study, professionals who engaged in Fused Co-Creation (Cyborgs) newskilled themselves in generative AI-related capabilities. Professionals who engaged in Abdicated Co-Creation (Self-Automators) failed to gain new skills during human-AI collaboration, neither domain/task-related, nor GenAI related.

We found the human-AI co-creation perspective and the skill-building perspective to be particularly useful in our in-depth examination of human-AI collaboration in the BCG experiment. Our analysis of human-AI co-creation and skill-building contributes to the current literature on human-AI collaboration in three important ways. First, we elaborate how human actors may accomplish human-AI co-creation using three primary modes of engagement and corresponding types of human-AI interaction: *Fused Co-Creation (Cyborgs)*, *Directed Co-Creation (Centaurs)*, and *Abdicated Co-Creation (Self-Automators)*. Second, we show how the use of each form of co-creation facilitates different skill-building

processes. Thirdly, we highlight a new barrier to human-AI collaboration. Finally, we contribute to the heated debate of the future of work and expertise and human-machine relationships more broadly.

METHODS

Research Setting

We conducted a field experiment in collaboration with a leading global management consulting firm to assess how junior associates and consultants used generative AI (GenAI) in problem-solving. We built a custom “ChatGPT-like” platform using OpenAI’s GPT-4 API (April 2023) to capture all participant prompts and system responses with timestamps. A total of 244 professionals were asked to recommend which of three brands in a fictional company should receive strategic investment, drawing on interview notes with (fictitious) executives and historical business performance data. Participants were instructed to submit a CEO memo (≤ 500 words). Submissions were scored for correctness (1/0; men’s brand was correct) and persuasiveness (1–10).

Data Collection

Activity logs. We collected 244 full conversational logs, comprising 4,975 interactions. Activity logs documented how each professional interacted with GPT-4 across the workflow, providing granular evidence of human–AI collaboration. During interviews, researchers also used these logs to probe participants’ reasoning and choices.

Interviews. Our team of 10 qualitative researchers conducted 237 semi-structured interviews (Spradley, 1979), ~60 minutes each, soon after task completion. Interviews explored nine topics including engagement with AI, time pressure, trust, creativity, and professional identity. Protocols were iteratively refined through collective analysis (Glaser & Strauss, 1967; Strauss & Corbin, 1990). For instance, reframing questions about autonomy through the lens of team dynamics yielded richer responses. Midway, a deeper inductive analysis revealed patterns in problem-solving with GenAI, leading to five new protocol questions. Interviewees were also shown artifacts such as their own prompts or visual depictions of the problem-solving subtasks, eliciting detailed reflections. All interviews were recorded with consent and transcribed. Further methods in detail are in **Appendix B: Detailed Methods**.

Data Analysis

We followed an inductive theory-building process (Glaser & Strauss, 1967; Strauss & Corbin, 1990; Golden-Biddle & Locke, 2007).

Phase 1: Identifying subtasks. We reviewed all prompts (n=4,975), which led us to classify human–AI interaction into seven subtasks (**Appendix A, Table 1: Business Analytics Subtasks**): framing, analyzing data (qualitative and quantitative), recommending, iterating, extending, evaluating, and reporting. For example, “Summarize the interview” was coded as analysis, while “Draft a CEO recommendation” was coded as reporting.

Phase 2: Coding practices. Using open coding (Charmaz, 2014), we inductively generated 14 practice codes (e.g., validating, mapping, modularizing, editorializing, pushing back; **Appendix A, Table 2: Human-GenAI Interaction Practices**). These practices captured the interaction patterns professionals used across subtasks.

Phase 3: Mapping practices and deriving co-creation types. We mapped practices to subtasks (Langley, 1999), supplemented by frequency tables to highlight prevalence across participants. Through iterative comparison, what emerged were three consistent co-creation types: Fused, Directed, and Abdicated Co-Creation. Each type reflected distinct distributions of practices across the workflow. *Phase 4: Integrating interviews.* Interview transcripts were analyzed inductively to identify how participants’ perceptions aligned with co-creation modes. This triangulation illuminated the logics underpinning participants’ choices.

Phase 5: Linking to skilling. Finally, we examined patterns of expertise-skill development across co-creation modes. Directed Co-Creation was associated with domain-specific upskilling, Fused Co-Creation with AI-related newskilling, and Abdicated Co-Creation with no skilling.

FINDINGS

In this study we closely analyzed how 244 professionals, BCG consultants, worked with GenAI to solve the same set of problems. Professionals were provided interview transcripts from company executives, historical financial data, and market information, materials that mirrored the data-rich,

ambiguity-laden environment of real consulting work. The task replicated a classic strategic consulting challenge: diagnosing performance issues and recommending a growth strategy for a portfolio of brands, a core activity through which consultants demonstrate analytical rigor, business judgment, and communication skill.

As noted earlier, our detailed analysis revealed three distinct modes of human–AI co-creation: Fused, Directed, and Abdicated, and three corresponding human-AI interaction types of professionals who enacted them: Cyborgs, Centaurs, and Self-Automators. While the modes capture the process of collaboration between human and AI, the types represent the behavioral patterns that emerged from those processes. Together, these findings reveal that the crucial story of GenAI in knowledge work is not only about *whether* professionals use AI, but *how* they do so. The way professionals integrate, direct, or abdicate collaboration with AI may have profound consequences for the problem-solving process, the skills they develop, and, as we will discuss later, the capabilities organizations build.

Our analysis revealed the three hidden patterns that were distributed unevenly across the professionals in our study. The majority, 60% (N=146), engaged in Fused Co-Creation (Cyborgs), integrating GenAI continuously throughout all stages of problem-solving. Their collaboration unfolded as an iterative dialogue: probing AI outputs, extending ideas, and validating results in a seamless rhythm of joint problem-solving.

Another 14% (N=34) practiced Directed Co-Creation (Centaurs), using GenAI selectively for specific subtasks like mapping, gathering information, or refining content, while maintaining control over the overall analytical logic. These professionals remained firmly in the driver's seat, leveraging GenAI to enhance efficiency and polish outputs without surrendering their judgment.

The remaining 27% (N=63) engaged in Abdicated Co-Creation (Self-Automators), relying heavily on GenAI in later stages, particularly for reporting, and delegating both analytical and evaluative thinking to the system. Their work was fast and polished but lacked depth, resembling outputs completed for them rather than with them. **Table 1: Modes of Human–AI Co-Creation in Problem-Solving and Corresponding Types** summarizes the defining features of these three modes and types, including their

interaction patterns, representative practices, and distribution across the sample.

Table 1: Modes of Human–AI Co-Creation in Problem-Solving and Corresponding Types

Co-Creation Mode	Individual Interaction Type	Interaction Example
Fused Co-Creation	Cyborg	Professional (Recommendation and Extending Subtasks): Lets zoom into the first issue you identified, which is the strategic mistake where Kleding Man shifted its target customer to younger, more fashionable individuals. How will you remedy this? You are a management consultant working for Kelding, an apparel company. In a previous session, you obtained the following information:... [INTERVIEW DATA and QUALITATIVE DATA]..Based on this, which segment (men's, women's or children's should we target? GenAI (Recommendation and Extending Subtasks): Based on the market trends and Kelding's market share in each segment, it would be strategic to initially target the womenswear segment. The primary reason is that Kelding's market share in womenswear has remained relatively stable and even increased slightly in recent years despite the overall decline in womenswear market share in the broader market. This indicates that Kelding Woman has a positive brand image and potential for growth... Professional (Iterating Subtask): However, you have missed the information that the men's section is growing rapidly. We would prefer to target the men's section. Below are the problems you identified with the men's section: Based on the interviews with Daren F. and Emma F., the key reasons for the decline in Kelding's menswear market share, despite the broader market trend, can be attributed to: 1.Strategic Mistakes: Daren F. highlighted a strategic mistake... GenAI (Iterating Subtask): To turn the men's section around, consider the following strategic recommendations: 1. Refocus on original target customers:... Professional (Writing Subtask): Please prepare a 500-750 word note to the CEO. The note should focus on the following: If the CEO must pick one brand (men's, women's or children's) to focus on and invest to drive revenue growth in the company, what brand should that be?... GenAI (Writing Subtask): Subject: Recommendation on Brand Focus and Action Plan for Revenue Growth Dear [CEO's Name],... Recommendation: Focus on Kelding Woman... Professional (Iterating Subtask): I would like to change the recommendation to focusing on the men's section. Please rewrite the above, and mention the following points: 1) The reason the men's section is facing declining revenues 2) It is important to focus on the men's market, since it is growing rapidly

		3) How focusing on the men's segment will have spill over effects on the kids's segment due to its close association with the kids's segment 4) The strategies we can use to increase revenue in the men's section GenAI (Iterating Subtask): Subject: Recommendation on Brand Focus and Action Plan for Revenue Growth Dear [CEO's Name],...Recommendation: Focus on Kelding Men...
Directed Co-Creation	Centaur	Professional (Data Analysis Subtask): What are the customer segments in male dutch clothing? GenAI (Data Analysis Subtask): In the male Dutch clothing market, customer segments can be broadly categorized based on factors such as age, income, lifestyle, and fashion preferences. Here are some of the primary customer segments: 1. Teens and young adults (15-24 years old):...
Abdicated Co-Creation	Self-Automator	Professional (Entire Problem-Solving Workflow): The CEO, Harold Van Muylders, would like to understand Kleding's performance by the company's three brands (Kleding Man, Kleding Woman, and Kleding Kids) to uncover deeper issues. Please find attached interviews from company insiders on this issue. In addition, the attached Excel sheet provides financial data broken down by brands. GenAI (Entire Problem-Solving Workflow): Subject: Recommendation for Focusing on Kleding Woman Dear Mr. Van Muylders,...

These were not isolated anecdotes. In our field study of 244 professionals, we observed the same three modes unfolding across a structured, seven-stage problem-solving workflow. Though every professional faced the same task and had access to the same tool, their choices about when to engage GenAI and how to engage with it led to meaningful differences in practice.

To capture these patterns, we developed the **Figure 1: Human-AI Knowledge Co-Creation Framework**, a conceptual model for understanding how professionals integrate GenAI into complex, knowledge-intensive work. The framework is built around two fundamental questions that structure any collaborative problem-solving dynamic between human and machine:

- **Who selects what needs to be done?** — whether the human or GenAI defines the work and sets the workflow agenda.
- **Who identifies how it gets done?** — whether the human or GenAI determines the division of labor, the work methods, logic, or content used to carry it out.

Crossing these two dimensions produces a 2×2 typology that reveals the architecture of human–AI collaboration modes, not just what GenAI is used for, but who is steering the workflow, how the work gets actually divided and how the human expert’s skills and knowledge are being used or not while working with GenAI in practice.

Figure 1: Human-AI Knowledge Co-Creation Framework

		Human-driven (What task is done)	AI-driven (What task is done)
Human-driven (How the task is done)		Centaur (Directed Co-Creation)	Unnamed
	AI-driven (How the task is done)	Cyborg (Fused Co-Creation)	Self-Automator (Abdicated Co-Creation)

Figure 1 maps human–AI interaction styles across two dimensions: whether task direction is primarily driven by the human or the AI (horizontal), and whether task execution is primarily carried out by the human or the AI (vertical). The three empirically grounded modes, Fused (Cyborg), Directed (Centaur), and Abdicated (Self-Automator) Co-Creation, each occupy a unique quadrant, capturing how control flows between human and AI in real-world settings. The fourth quadrant, where AI drives task selection but humans drive execution, remained empirically empty in our study. This absence suggests that when professionals cede control over *what* to work on, they rarely retain control over *how* to do it. The cognitive and practical demands of executing work that one did not define may create friction that pushes professionals toward full self-automation.

Professionals’ Perceptions Shaped Their Adoption of Different Co-creation Modes and Interaction Types

What’s striking is that although every consultant had access to the same tools and the same task, their choices, about when to engage GenAI and how much authority to give it, reflected deeper assumptions about their own role and the role of technology. These assumptions, in turn, had significant implications not just for task performance, but for skilling, shaping how professionals developed expertise and AI-related capabilities (or not) as they engaged with GenAI. We dive into each of the three modes below, beginning with Fused Co-Creation, where human and AI work in lockstep to shape both the problem and its solution.

Fused Co-Creation (Cyborgs). Those engaging Fused Co-Creation engaged intensively with GenAI, blurring the lines between AI and human interaction. This approach involved continuous iteration, directing AI to conduct analyses, questioning its outputs, and refining solutions in a dynamic, back-and-forth manner. By leveraging AI as an interactive tool, they maintained active involvement in shaping both the process and the final solution. As Aisha Hassan explained, “I was kind of guided in my second answer to give a different response through that conversation [with GenAI]. Probably a better response.”

Professionals who engaged in Fused Co-Creation were influenced by two key factors: perceived productivity gains and a willingness to develop AI-related expertise. For example, Lucky Vaughan

shared, “It was definitely faster. Generally, it was a huge help in summarizing data...and in terms of how to structure it all together... For drafting the recommendation, it’s super helpful because generative AI took all the data while I skimmed through.” Others noted how GenAI reduced their dependency on colleagues, as Mei Chen stated, “It might reduce dependency on others for sound boarding. Typically associates consultants talk among themselves before going on to the higher level... but at least at the very lowest level sound boarding might be outsourced to Chat GPT.”

The balance of control was central in Fused Co-Creation. Professionals found that AI could produce multiple compelling outputs, but human intervention was still needed to finalize the best answer. As Diego Alvarez described, “I was actually able to get the AI to generate two different answers and it had compelling reasons to believe both so then it really took the human to step in and say, okay, which ends which answer really doesn't make a lot more sense.”

Directed Co-Creation (Centaur). Those engaging in Directed Co-Creation took a more selective approach. Instead of relying on AI throughout, they activated their own judgment first, leveraging AI to enhance their domain knowledge and support specific phases of the problem-solving process. They maintained a structured, controlled interaction with AI, using it as a tool for targeted assistance rather than continuous engagement. Many professionals saw AI as an augmentative tool rather than a collaborator, as Sara Haddad shared: “I think of it as like a tool in a toolbox. It's something that I can pull out when I need but it's not like my smartphone, that's like always on me in my pocket. I use it [AI] selectively for specific tasks but I don't think of it as like a ubiquitous item.”

This selective use underscored a desire to maintain control and avoid skill degradation. For example, Jonas Bergström remarked, “It's like the automatic spelling. You lose that capacity to actually think by yourself, so that's what I was scared about.” Others resisted over-reliance on AI, fearing it could undermine their own expertise. Priya Iyer explained, “Trying to cheat when it comes to problem-solving... I think I need to be careful because at this level most of the learning should come from me. I should be actively learning and should not be so reliant on it so soon.”

Abdicated Co-Creation (Self-Automators). Finally, professionals engaging in Abdicated Co-Creation fully delegated the problem-solving process to AI. After providing AI with the task and supporting data, they neither iterated nor actively engaged in shaping the solution. This approach allowed for rapid outputs. As Mateusz Kowalski put it, “I’ve been really more productive with GenAI than without it.” However, it also risked losing the depth of human expertise and strategic thinking.

For some, full delegation stemmed from trust in AI’s capabilities. Leila Mansour described their approach: “I just basically copy and pasted the data into the prompt. It gave me something that seemed okay. I did a quick sanity check, just to check that the data and the recommendation was not too far from reality.” Others valued AI’s efficiency in handling tasks they found tedious. As Tomoko Sato candidly admitted, “So there I was without AI and I was feeling lazy. Frankly, I was like, how should I write 500 words now? I don’t want to write 500 words, I want GenAI to produce it.” Ultimately, Abdicated Co-Creation reflected a profound reliance on AI for task execution. Professionals saw it as a means to maximize their productivity while allowing themselves to step back, confident in the machine’s ability to deliver results.

These varying levels of interaction reflect broader implications for professional development. As AI continues to evolve, professionals must consider whether they are simply using AI to increase their productivity or actively co-creating with it to enhance their capabilities.

Operationalizing Co-Creation: Patterns of Interaction Across Modes

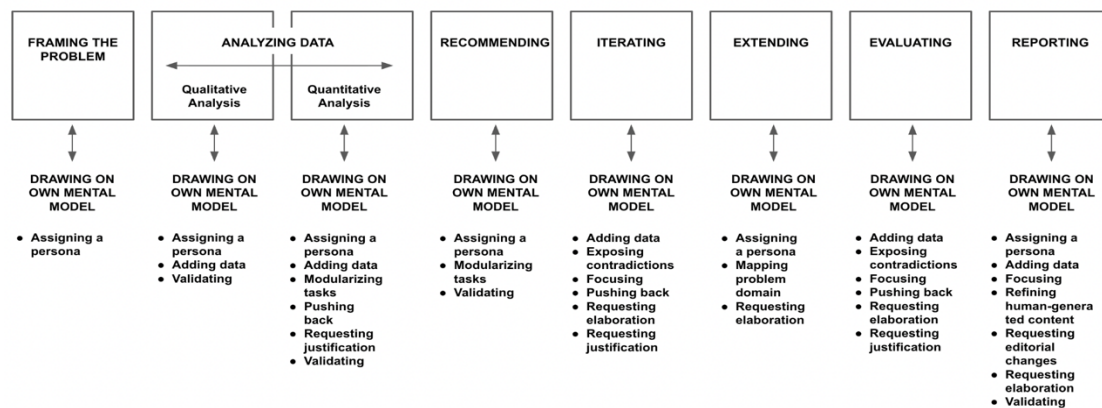
While the previous section outlined why professionals adopted different co-creation modes, based on how they distributed decision-making authority with AI, this section examines how those modes were enacted in practice. Specifically, it details the ways professionals interacted with GenAI across different phases of the problem-solving workflow, beginning with those engaged in Fused Co-Creation.

Fused Co-Creation (Cyborgs). The approach taken by professionals engaged in Fused Co-Creation was evident in the ways they interacted with GenAI. Their engagement was both structured and exploratory, combining established prompting best practices with conversational experimentation. This flexible method enabled them to explore the capabilities of AI while maintaining control over the process.

Those engaging in this mode consistently used AI across all phases of the problem-solving workflow: framing the problem, analyzing data, generating recommendations, iterating on outputs, extending ideas, evaluating results, and drafting reports. Rather than using AI as a passive tool, these professionals interacted with it in ways that shaped the process itself, with AI outputs informing and influencing each stage of their work.

Of the 244 professionals, 60% (N=146/244) engaged in Fused Co-Creation during the problem-solving process (**Appendix A, Table 2: Human-GenAI Interaction Practices**). Their intense engagement wasn't just about efficiency; it was about unlocking new ways of thinking and problem-solving. This deep integration led to the use of diverse practices throughout the task, (**Figure 2: Subtasks and Fused (Cyborg) Co-Creation Practices**) showcasing how Fused Co-Creation transformed not only the outputs but the very process of collaboration itself.

Figure 2: Subtasks and Fused (Cyborg) Co-Creation Practices



Problem Framing. When framing the problem and analyzing data, those engaged in Fused Co-Creation assigned a persona to GenAI. The persona practice is a known best practice where GenAI is instructed to take on a role and simulate a specific type of personality or character. Used in the framing phase, professionals assigned a persona to add context the machine lacked to the problem space. For example, Kwame Mensah instructed GenAI to act as a consultant, prompting, “Imagine that you are a consultant in a strategy consulting firm. You are working for a client called Kleding...” Kwame Mensah orients GenAI

and then proceeds to the next subtask, prompting, “The interview notes are hereafter. Could you summarize their content in 500 words?” Generally, those engaging in Fused Co-Creation combined framing the problem and data analysis phases.

Analyzing Data and Recommending. In the analyzing data and recommending phases, professionals engaged in practices to reduce errors, such as the practices of modularizing and validation. Professionals used the practice of modularizing, where work is broken down into multiple sub-steps for GenAI to execute. This best practice is also known as “chain of thought prompting,” often used to reduce the complexity of a task and the likelihood of GenAI becoming confused. Professionals used this practice primarily in the data analyzing and recommendation phases; phases that were more complex with multiple steps when compared to other phases. For example, Petrova instructed GenAI to take multiple steps, first for AI to review the facts, second to conduct data analysis, and third, using the persona practice, to redo the analysis and provide the recommendation from the perspective of a senior consultant, “Provide recommendations to solve the problem. Following these steps: 1. Provide an overview of the facts to consider, 2. Provide the analysis, 3. Redo the analysis with additional accuracy. Please give the recommendations from the perspective of a senior business consultant.” GenAI responded, providing an output that progressed through each of the steps listed by Petrova. Through modularizing, professionals engaging in Fused Co-Creation maintained a dynamic balance with GenAI, structuring the workflow while allowing the system to conduct the data analysis and recommendation development.

Professionals often used the validating practice when engaging with GenAI in the analysis and recommendation subtasks to push the system toward greater accuracy. Validating involved asking GenAI to check its own inputs, analysis, or outputs. Unlike professionals in Abdicated Co-Creation, who passively accepted GenAI’s outputs, those in Fused Co-Creation frequently prompted it to review its work, sometimes multiple times. It was used to check the reasoning behind GenAI’s recommendations, often after it produced an initial output. For instance, Felix Schneider received a recommendation from GenAI to not focus on the Kleding Man or Kleding Kids brand after conducting the qualitative and quantitative analysis, and then immediately directed GenAI to verify the result: “Check the accuracy of

this answer.” GenAI responded by confirming its earlier output, stating, “The answer accurately reflects the reasons for not selecting Kleding Man and Kleding Kids as the focus, based on the information provided earlier...” It further justified its position by referencing prior data: “Moreover, it includes the critical statistic about 80% of sales to older men being linked to their children's shopping...” Satisfied with the reasoning, Schneider accepted the output and advanced to the next phase of the problem-solving workflow. Unfortunately for this professional, the correct answer was Kleding Men’s, not Kleding Women’s. Schneider had employed a best practice of validating and was persuaded by GenAI even when GenAI had given the incorrect answer. This happened to many professionals using GenAI in the Fused Co-Creation mode, as we discuss below.

Iterating. In the iterating phase, professionals engaging in Fused Co-Creation relied on frequent exchanges with GenAI to refine and extend their analyses. They experimented by providing additional information, questioning outputs, asking GenAI to focus on particular aspects, or explicitly disagreeing with its conclusions. One of the most common practices in this phase was *adding data*, in which professionals introduced new inputs after GenAI produced an output, prompting it to rerun its analysis. This iterative exchange reflected a form of conversational experimentation, where professionals contributed information while GenAI drove the analytical reasoning.

For instance, after receiving a recommendation for the women’s brand based on interview and financial data, Amara Okafor added market data and asked, “Based on the financial data the overall market for menswear looks to have grown about 72% from 2013 to 2017. Does that change which brand the CEO should focus on?” GenAI reconsidered and replied, “Considering the overall market for Menswear has grown about 72% from 2013 to 2017, the CEO should reevaluate the focus on Kleding Woman and also consider investing in Kleding Man...” By strategically introducing new data, Okafor prompted GenAI to reconsider its analysis and refine the reasoning behind its recommendation.

Another practice that was heavily used in the iterating phase was the practice of pushing back, disagreeing with the output, and asking AI to reconsider. Unlike the validating practice, when pushing back, professionals were not directing GenAI to check its work but instead directly disagreeing with it

and asking it to revisit its decision. For example, Hiro Tanaka received the initial recommendation of women's brand from GenAI, to which they disagreed and suggested that the kids brand may be the recommendation to consider:

I don't think I agree with your conclusion. The women's brand is doing well, so I don't think it needs attention. I would instead recommend focusing on Kleding Kids.... Can you rewrite the above around this recommendation?

GenAI revised its response accordingly. Yet upon review of the new output, Tanaka pushed back again, changing their mind on the recommendation, telling GenAI that the recommendation should be the men's brand, stating:

I've changed my mind. I think that Kleding Kids is probably too small of the total market share and too small of Kleding's revenue. Please rewrite with the recommendation being a focus on Kleding Men. It is the growing part of the total market...

The result of pushing back a second time was a revised recommendation. When using the pushing back practice it often resulted in multiple iterations of an output as GenAI is questioned, disagreed with and is sometimes in conflict with what the user believes to be the correct answer.

Professionals also used several other practices in the iterating phase that resulted in exploratory and interactive conversations with GenAI: requesting elaboration (asking GenAI to bring more breadth of details and nuance on an interesting or unexpected point), requesting justification (asking GenAI to explain a confusing output, or why a particular recommendation was made), exposing contradictions (pointing out logical or factual inconsistencies), and focusing (directing GenAI to focus on a particular data point, content, or task).

Extending, Evaluating and Reporting. Those engaged in Fused Co-Creation did not interact with GenAI much in the extending and evaluating phases. They drew on GenAI heavily in the solution reporting phase, requesting editorial changes to GenAI output (high number of interactions) but also assigning GenAI a persona for writing the memo to the CEO (medium number of interactions). For example, one professional Isabel Sousa directed GenAI to answer the exact question posed by the

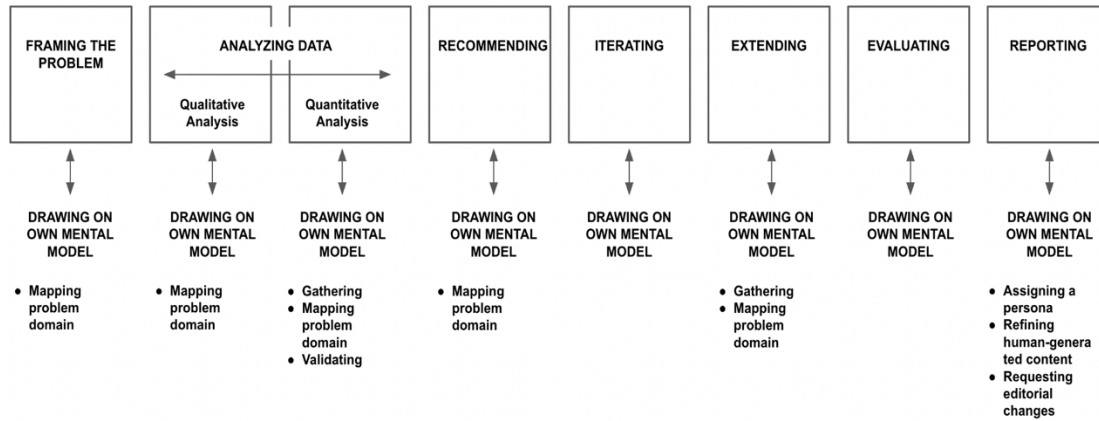
problem-solving task (Please prepare a 500 - 700 word note to the CEO...), but tailoring this to their own solution, saying:

Great. Please rewrite the email and remove “I hope this email finds you.” Begin the sentence with the recommended action. Please remove any direct quotes from Emma F. and just include a one sentence summary....

Those engaging in Fused Co-Creation often made editorial changes with GenAI as they collaboratively refined the final memo in the reporting phase of the problem-solving task. Examples of each of these practices and their intensity are shown in **Appendix A, Table 2: Human-GenAI Interaction Practices** and **Appendix A, Figure 1: Fused (Cyborg) Co-Creation Practice Map**.

Directed Co-Creation (Centaur). Professionals who engaged in Directed Co-Creation approached GenAI with a measured and selective strategy. Their interactions were characterized by targeted use of prompting practices aimed at supporting specific subtasks, while maintaining control over the overall problem-solving workflow. Unlike those in Fused Co-Creation, these professionals did not integrate AI throughout the workflow but instead interacted with it during particular phases. As shown in **Figure 3: Subtasks and Directed (Centaur) Co-Creation Practices**, their engagement was concentrated and deliberate, reflecting a clear intent to preserve task ownership. They primarily relied on three prompting practices: mapping the problem domain, gathering information about methods, and refining content they had generated themselves (**Appendix A, Table 2: Human-GenAI Interaction Practices**). Of the 244 professionals in the study, 14% (N=34) exhibited this pattern of Directed Co-Creation.

Figure 3: Subtasks and Directed (Centaur) Co-Creation Practices



Problem Framing. Those who engaged in Directed Co-Creation had stark differences in how they interacted with GenAI when compared to professionals using Fused or Abdicated Co-Creation modes. To frame the problem, unlike those engaged in Fused Co-Creation, who had assigned a persona to first orient the AI before starting the next phase, those engaged in Directed Co-Creation asked GenAI for assistance in mapping the problem domain to help them understand and define the problem, explore potential goals and success criteria of the solution generation process, and the steps required to accomplish these goals. Mapping the problem domain entailed asking GenAI for general information related to the problem’s domain for the human to use to better understand the problem. For example, those engaged in Directed Co-Creation mapped the problem domain by asking GenAI, “What are examples of clothing retailers that turned around a men's clothing brand after poor performance?” or “Can you provide an example of a menswear clothing retailer that improved its in-store experience to drive revenue growth?”

Analyzing Data and Recommending. In the phases of analyzing data and recommending, professionals turned to GenAI for mapping the problem domain and gathering methods information, each of which served as an alternative to using an online search engine like Google. Here professionals used the mapping practice in the qualitative analysis to ask questions to assist in their data analysis like Omar El-Sayed, who asked, “What are key trends in menswear?”

In contrast to the mapping practice, which was centered on providing insights into the problem's domain, those engaged in Directed Co-Creation used the gathering practice to ask GenAI for specific

information on methods that the human is employing to solve their sub-task. For example, Jiwoo Paik asked, “How do I calculate the market size growth rate of some industry from 2013 to 2017?” GenAI responds to Paik, “To calculate the market size growth rate of an industry between 2013 and 2017, follow these steps:...” and provides four steps to follow, including the formula for the compound annual growth rate (CAGR). Upon receiving the step-by-step process in calculating market size growth, Paik follows up their query asking GenAI for additional information to help them in their quantitative analysis, now asking for a formula to use in Excel, prompting, “What’s the Excel formula for this?”, to which GenAI provides the formula with example for CAGR to Paik. While those engaging in Fused Co-Creation would then ask the GenAI to run that calculation and complete the work in the analysis phase, professionals engaging in Directed Co-Creation took the information provided by GenAI and did the work themselves. In this example, Paik took the Excel formula and did the analysis in Excel themselves, relying on their own abilities and not GenAI. Unlike those engaging in Fused or Abdicated Co-Creation, professionals primarily used GenAI in the data analysis and recommendation phase to broaden their own knowledge and speed up their learning curve in a specific context.

Iterating. In the iterating phase, professionals engaged in Directed Co-Creation did not turn to GenAI at all. Instead, they iterated by relying on their own understanding of the task and their skills rather than interacting with GenAI.

Extending. In the extending phase, professionals engaged in Directed Co-Creation turned to GenAI to help extend their insights while those engaged in Fused Co-Creation drew on GenAI less in this phase and those engaged in Abdicated Co-Creation did not draw on GenAI at all. In the extending phase, professionals engaged in Directed Co-Creation used the prompting practice of mapping the problem domain to inform themselves in the development of innovative tactics, for example, Ananya Patel asked, “What are ways Tommy Hilfiger increased brand awareness among men?” Patel followed up using the practice of gathering information to use GenAI to expand on the methods, “How can a clothing retailer reposition its price to resonate with a younger male target segment?” Like previous interactions,

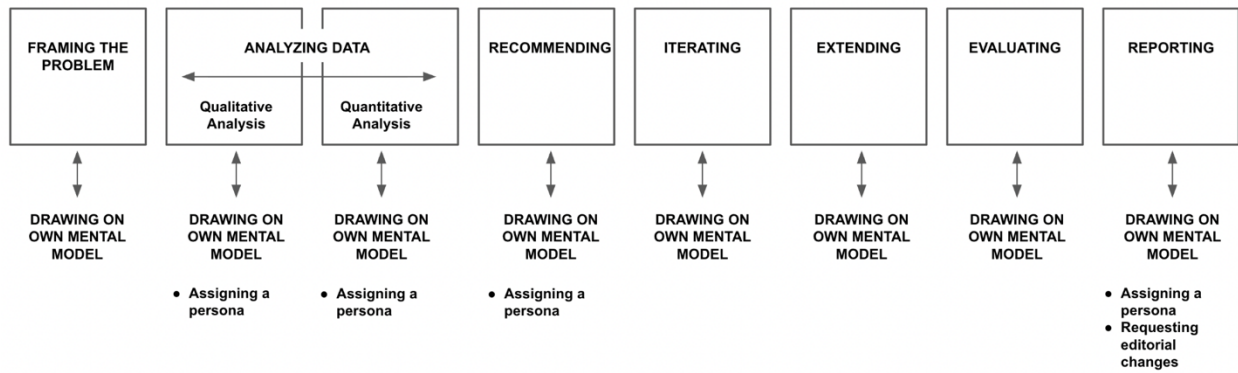
professionals were leveraging GenAI to build upon their own skills, in this case, to inform in the ideation process in the extending phase.

Evaluating. To evaluate the solutions, those engaging in Directed Co-Creation did this on their own, without asking for assistance from GenAI.

Reporting. When reporting solutions, those engaged in Directed Co-Creation drew on GenAI to refine their own human-generated-content and to request editorial changes. Unlike those engaged in Fused Co-Creation who worked with GenAI in reporting, developing an initial draft together and further editing, those engaging in Directed Co-Creation did not begin with an AI output. Instead, they brought their own solutions to the interaction, and asked AI for help in refining these self-generated solutions. For example, one professional, Mariana Costa asked: “Will you please review this for ease of reading, conciseness, and overall quality?...” By using GenAI to refine a draft, bullet points, or notes into a polished piece of writing, professionals maintained control over the core content and meaning of the text, employing GenAI to improve the superficial and structural aspects of the writing. Examples of each of these practices and their intensity are shown in **Appendix A, Table 2: Human-GenAI Interaction Practices** and **Appendix A, Figure 2: Directed (Centaur) Co-Creation Practice Map**.

Abdicated Co-Creation (Self-Automators). Professionals who engaged in Abdicated Co-Creation demonstrated a markedly lower level of interaction with GenAI compared to those in the Fused or Directed Co-Creation mode. Their engagement was not distributed across the full problem-solving workflow but was instead largely concentrated in the reporting phase. Of the 244 professionals in the study, 27% (N=63) exhibited this mode of engagement. Among them, 44% (28 of 63) fully automated the task, accepting the AI’s output without modification, while the remaining 56% (35 of 63) made only superficial edits to the AI-generated content (**Figure 4: Subtasks and Abdicated (Self-Automator) Co-Creation Practices**). This limited interaction suggests a shift in how these professionals approached the task, delegating not only execution but also evaluative judgment to the AI.

Figure 4: Subtasks and Abdicated (Self-Automator) Co-Creation Practices



Unlike those employing a Fused Co-Creation and Directed Co-Creation mode, those engaging in Abdicated Co-Creation consolidated multiple phases of the problem-solving workflow into one or two interactions. Consolidation was a result of professionals copying and pasting the entire problem statement, interview and financial data into a single prompt in the qualitative analysis phase of the problem-solving process. Upon receiving everything, GenAI then responded with an output which the professional accepted, with the exception of making small editorial changes in the reporting phase. Professionals engaging in Abdicated Co-Creation allowed GenAI to do all the work, almost fully removing themselves and their own understanding of the situation and skillset from the process. For example, Carlos Mendez, directed GenAI, “Please consider the following data first,” and then copied and pasted all interviews, and all financial tables into GenAI. GenAI responded, providing Mendez with a summary and highlights of the analysis it conducted, “Based on the provided information, Kleding’s financial data, market data and interviews with key personnel highlight the following insights...” Mendez then immediately input the prompt that was provided as part of task, asking GenAI for the recommendation and simultaneously moving to the extending, evaluating and reporting subtask. Mendez prompted, “...What brand should that be?...,” text associated with the recommendation subtask, then continued their prompt to include, “What is the rationale for the choice?,” text associated with the evaluating subtask and also included, “Please also suggest innovative and tactical actions the CEO can take to improve this chosen brand to drive revenue growth...,” text associated with the extending task.

Mendez concluded the problem-solving task by directing GenAI to report the output, including in the same prompt as the previous text, “Please prepare a 500 – 750 word note to the CEO.”

In sum, Mendez completed the problem-solving process with minimal interaction, strictly following the instructions provided to them without challenging GenAI’s outputs. This stood out because these were skilled professionals who took the task seriously, knowing their performance would be noted in their annual review.

Reporting. While Mendez did not draw on their own understanding and skill to push back on GenAI at all, in some cases of Abdicated Co-Creation, professionals interacted with GenAI slightly more in the refinement of their outputs before accepting a final version. They used AI to make superficial, structural or grammatical changes to the reporting; they did not use GenAI to change core elements of the output, e.g. changing the recommendation. For example, one professional made changes to the outputs that AI had produced, directing: “I like number 2 and a combination of 4 and 7. Can you rewrite this shorter list?” Professionals employing the Abdicated Co-Creation mode of engagement used editorialization practices to refine outputs but generally limited their interaction, relying on GenAI to solve the task. Examples of each of these practices and their intensity are shown in **Appendix A, Table 2: Human-GenAI Interaction Practices** and **Appendix A, Figure 3: Abdicated (Self-Automator) Co-Creation Practice Map.**

Implications of the Co-Creation Mode and Type on Skilling

The mode of human-GenAI co-creation significantly influenced the impact on skilling for professionals, shaping how they interacted with both their tasks and AI itself. We identified three ways skilling was impacted with the use of GenAI: upskilling, where users developed their task-related expertise; newskilling, where users developed their AI-related expertise; and no skilling, where users did not develop either task-related or AI-related expertise (**Table 2: Skilling Impact**).

Table 2: Skilling Impact

User Type	Expertise Development
Fused Co-Creation (Cyborgs)	• Newskilling: Development of AI-related expertise • Maintenance of task-related expertise
Directed Co-Creation (Centaurus)	• Upskilling: Development of task-related expertise
Abdicated Co-Creation (Self-Automators)	• No skilling: No development of task-related and AI-related expertise

Newskilling. In Fused Co-Creation, the professional (Cyborgs) and GenAI engaged in an iterative process, exchanging information and adapting to each other's inputs. This back-and-forth refinement is a key feature of Fused Co-Creation, impacting skilling in two ways. First, the professional developed new GenAI-related expertise by experimenting with the interaction and adapting their decision-making based on AI inputs and outputs. Second, at the same time, they maintained their task-related expertise by continuously responding to, checking, and validating the AI's outputs.

Professionals in Fused Co-Creation adapted their interactions with GenAI through experimentation, acquiring new expertise in the process (see **Appendix A, Table 3: Fused Co-Creation Newskilling Examples**). For instance, Sofia Rossi described how they adjusted their interactions with GenAI as they developed new AI-related skills:

... For my own work recently, I was treating generative AI much more conversational, so being able to push back and dig deeper and keep asking many, many follow up questions. As opposed to before I would just ask it the first question and then just take that as a starting point and not think that I had the ability to dive deeper...

Others reflected on how they newskilled by refining how they communicated with GenAI, as they felt it was different when compared to communicating with an associate. For example, Yara Ben-David, shared:

I think a key challenge is communicating with it is the specificity of the tasks that I want to communicate with it. When I talk to it, I give a comment and criticize its work and tell it ‘Hey, come back to me with that’. That was a bit hard. If I compare ChatGPT to a junior associate or an associate when I’m working with a team and I assigned him to do something it’s easier for me to tell the associate ‘No no do this, do that’ and I pride myself on being super clear in my instructions. Yet it was a bit of a challenge for me to explain to GPT like, ‘Hey, change this to that.’

Second, those interacting in Fused Co-Creation maintained their task-related expertise which was evidenced in the use of practices such as explanation, exposing and pushing back, as the professional remained in the loop and fully engaged in the workflow. For example, Rohan Gupta responded to GenAI’s initial response of, “Kleding Woman poses the most absolute revenue increase potential...,” after conducting data analysis by pushing back, as they had noticed potential missteps in the analysis, stating:

Why don't you think Kleding Man has potential? The company lost market share due to a bad strategic choice, but they could gain that share back if they revert that choice and fix other issues, which is what Daren F. plans to do. Considering this, and considering the Men's market is by far the largest of the 3 markets, do you not think Kleding stands to gain substantial revenue from the men's segment if they implement the necessary changes?

In response to Gupta’s input, GenAI affirmed that Gupta was correct in that, “there is potential for Kleding Man to gain market share and revenue if they successfully implement the necessary strategic changes...Kleding Man could benefit from targeting this larger market effectively.” By staying engaged in the process, Gupta did not cede their expertise.

In another example, after receiving the recommendation to invest in the women’s brand, Elena Popescu exposed a factual inconsistency in its rationale for the selection, as GenAI noted that, “The women’s brand...already showing consistent growth with a 6% CAGR between 2013 and 2017....and the man’s brand is.... facing declining sales with a -5% CAGR between 2013 and 2017.” Remaining in the loop and exercising their expertise by identifying a factual inconsistency, Gupta responded, “The menswear market is growing at 12% CAGR while the womenswear market is growing at 5% CAGR.

Does this change your answer?”. GenAI responded, changing its recommendation to the men's brand, the correct answer.

When engaging in the Fused Co-Creation mode, professionals actively collaborated with GenAI, leveraging their own skills while developing new expertise by interacting and experimenting with AI's capabilities to achieve the desired outcome.

Upskilling. While professionals in Fused Co-Creation newskilled, those in Directed Co-Creation (Centaurs) upskilled by developing their task-related expertise through interaction with GenAI. For example, Priya Iyer described using GenAI to improve their task-related expertise:

I still went to the data set and did calculations but asking GPT for formulas was a lot more convenient and it would actually give you an example. To me it's a one-stop service. With ChatGPT I don't think I needed to go back to Google or any other search to find something else.

Miguel Santos shared that they used GenAI to improve their ideation and writing skills:

I was thinking, 'Okay, I'm going to do the bulk of the thinking and I will use GPT when I'm finalizing or I'm stuck with something... Given my expectations that GPT is good for wordsmithing, and is good for divergent thinking, I can use it in very targeted ways.

The limited and directed practices used by professionals illustrate how they perceived GenAI as a mechanism for developing their domain expertise (**Appendix A, Table 4: Directed Co-Creation Upskilling Examples**). For example, when learning about a new domain, such as retail clothing, Hana Novak, first asked, “Could you give me several examples of clothing retailers who have gone back to their original target market (mid-aged men) after unsuccessfully trying to target younger men?” GenAI responded by presenting two examples of organizations with descriptions, Dockers and Men's Warehouse and some analysis, stating:

Some brands and retailers have experienced varying degrees of success when diversifying or attempting to appeal to younger demographics...the key is to strike a balance between the brand's identity and customer preferences...

Using this feedback, Santos gained domain knowledge, improved their skills, and further explored Dockers' approach, prompting, “Great, thanks. Could you tell me more specifically how Dockers managed to shift its focus back on mid-aged men?”

GenAI’s initial response, citing examples from Dockers and Men’s Warehouse, directly influenced Santos’s upskilling by prompting further exploration of Dockers' actions. Santos used GenAI to gain domain knowledge, but the AI's feedback spurred additional learning, prompting Santos to push their own upskilling related to the task. In Directed Co-Creation, professionals used GenAI to upskill while doing the work themselves.

Yet, as they were upskilled in task-related functions, professionals who engaged in Directed Co-Creation did not acquire new AI-related expertise. Relying primarily on their own expertise, and limiting their own interaction with GenAI, rather than interacting fluidly, they failed to transform how they worked with GenAI as they did not experiment or fully integrate GenAI into their decision-making process. Various factors influenced professionals’ decision to not newskill, including: not being ready to make the change from how they currently worked, potential ethical considerations and wanting to maintain a deep understanding of the task they were working on (**Table 3: Directed Co-Creation (Centaur’s) Resistance to Newskilling**), for example, Priya Iyer explained:

...What is super important to me, is that I am able to learn a lot to the extent that...when I see something, I don't only see it from the surface but I see the real causes in the routine and the logic behind it, and that's someone that I want to become. But if I become the type of person that goes to Chat GPT for solutions, I might be able to learn a little bit by repeating asking for a solution many, many times but it wouldn't be as effective as if I thought it through and then ask GPT to just complement my answer.”

Table 3: Directed Co-Creation (Centaur’s) Resistance to Newskilling

Factor	Description	Illustrative Data
Not ready to change	Unsure about adopting new process due to uncertainty of AI-generated outcome	It's a new skill set you need to know how to use on a daily basis, and I'm not sure if I'm ready to make that extra step, just because the result is not guaranteed and so why would I change my usual pattern of doing things? - Fatima Al-Masri

Not ready to change	Unwilling to change process due to familiarity with existing	I don't think it was necessarily difficult to use but maybe my own mental barrier of using it, because I'm so ingrained in doing something a specific way and having the experience of doing a similar problem in my day-to-day work that I would just do by myself. - Liam O'Connor
Ethics	Concerned about getting credit for work that is not theirs	I still feel like the morality of having it craft the message for you, and then you using that as your own words, for me, it's still a bit tenuous, so pushing past that I think it's something that I'm still working on. - Gabriela Silva

Professionals who engaged in Abdicated (Self-Automators) Co-Creation had limited communication with GenAI. The minimal exchange of information regarding inputs, outputs, or decision-making processes resulted in no skilling, with no development of either task-related or AI-related expertise. As the user relied entirely on the AI to complete the task without engaging in mutual adjustment or learning from the AI's decision-making process, there was minimal change in the user's expertise.

Quality of Outputs

Each type of co-creation had different implications for the quality of professionals' work in our study. Those in Directed Co-Creation (Centaurs) had the highest accuracy in their business recommendations, outperforming those in Fused (Cyborgs) and Abdicated Co-Creation (Self-Automators). Additionally, professionals in both Directed and Fused Co-Creation (Centaurs and Cyborgs) excelled in persuasiveness, surpassing those in Abdicated Co-Creation.

In sum, the three types of co-creation had distinct impacts on expertise development. Fused Co-Creation (Cyborgs) developed AI-related expertise and maintained task-related expertise; Directed Co-Creation developed task-related expertise; and Abdicated Co-Creation did not develop either task-related or AI-related expertise.

DISCUSSION

Scholars of human-AI collaboration in professional settings have focused primarily on the interaction between domain expert professionals and predictive AI only at the point of recommendation,

detailing how most domain expert professionals have little ability to directly modify AI after development as they engage in problem-solving in their daily work (e.g., Anthony, 2021; Cameron, 2022; Cameron & Rahman, 2022; Pachidi et al., 2021; Rahman, 2021; Rahman & Cameron, 2022; Valentine & Hinds, 2022; Van den Broek et al., 2021; Waardenburg, Huysman & Sergeeva, 2022).

In this paper, we extend theories of human–AI collaboration beyond recommendation-level interactions to theorize how GenAI enables a continuous, conversational mode of co-creation. Our analysis shows that GenAI transforms human-AI collaboration from a single point of interaction into a continuous, conversational process. Because of GenAI’s conversational capabilities, professionals can weave it into every stage of problem-solving, framing, analysis, iteration, and reporting, making it a collaborator throughout, not just a recommender at the end. Recent empirical data from studies of GenAI suggest that professionals can interact iteratively with GenAI to accomplish various outcomes (Brynjolfsson et al., 2023; Csaszar, Ketkar & Kim, 2024; Dell’Acqua et al., 2023; 2025; Dillon et al., 2025; Noy & Zhang, 2023; Otis et al., 2023). However, these studies of GenAI do not theorize about the problem-solving process itself and the knowledge work collaboration and interaction between the professionals and AI; instead, they analyze performance implications.

We demonstrate how the concept of human-AI co-creation adds an important nuance to the current literature’s understanding of human-AI collaboration inside organizations.

Three Modes of Human-AI Co-Creation

Scholars of human-AI collaboration in professional settings have detailed how different groups of professionals, technology developers and domain experts, collaborate around AI primarily during the development process (Pachidi et al., 2021; Waardenburg, Huysman, & Sergeeva, 2022; Valentine & Hinds, 2022; Van den Broek et al., 2021). Once an algorithmic model is implemented in daily work, human-AI collaboration occurs by domain experts collaborating with one another around algorithms (e.g., Anthony, 2021; Rahman & Valentine, 2021; Karunakaran, 2022) or by accepting or rejecting algorithmic recommendations around specific tasks. This literature has elaborated human-AI collaboration processes such as police officers accepting or rejecting AI recommendations to patrol particular neighborhoods

(Brayne, 2017; Waardenburg, Huysman & Sergeeva, 2022), HR managers accepting or rejecting AI recommendations to hire particular candidates (Van den Broek, Sergeeva & Huysman, 2021), fashion buyers accepting or rejecting AI recommendations to buy particular articles of clothing to offer in their online stores (Valentine & Hinds, 2022), judges accepting or rejecting AI recommendations to provide bail to defendants (Christin, 2017), Uber drivers accepting or rejecting AI recommendations to accept rides (Cameron, 2022; 2023; Cameron & Rahman, 2022), and high-skill labor market platform workers accepting or rejecting AI recommendations to accept particular assignments (Rahman, 2021; Rahman & Valentine, 2021; Rahman, Karunakaran & Cameron, 2024). Once an AI model is out of development and implemented in daily work, the process of human-AI collaboration occurs around discrete tasks, at the point of recommendation, and entails a professional choosing to accept or reject the AI-informed recommendation.

In contrast, we find that, once a generative AI model is implemented in daily work, human-AI collaboration can occur around multiple different tasks, across multiple different points in a professional's workflow, and entails a professional engaging in a continuous, conversational mode of co-creation with GenAI. We further find that professionals may choose to engage in human-AI co-creation in one of three primary ways, which we call Fused Co-Creation (Cyborg), Directed Co-Creation (Centaur), and Abdicated Co-Creation (Self-Automator).

First, a professional may decide to engage in what we call "Fused Co-Creation" by relying on their own mental model of the current situation to determine what should be done, but relying on AI to determine how it should be done. In our setting, professionals who used Fused Co-Creation (Cyborg) tended to rely on AI to determine how things should be done in all phases of the human-LLM problem-solving process: framing the problem, analyzing data, recommending, iterating and extending solutions, evaluating solutions, and reporting solutions. They collaborated with AI by engaging in practices, such as: Adding data (adding data after an output is generated to re-do the analysis), Pushing Back (disagreeing with the output and ask AI to reconsider), Validating (asking AI to check its inputs, analysis, and

outputs), Explanation (Asking AI to explain a confusing output; or why a particular recommendation was made) and Exposing (pointing out logical or factual inconsistencies).

Second, a professional may engage in what we call “Directed Co-Creation” (Centaur) by relying predominantly on their own mental model of the current situation. We call this Directed Co-Creation because the human is the one who is determining both what should be done (drawing on their current mental model of similar situations to determine rules and guidelines for what to do) and how it should be done (drawing on their current mental model of similar situations to make choices of what to do). Participants who engaged in Directed Co-Creation tended to interact with GenAI to support or enhance their own task-related capabilities using three primary practices: mapping the problem domain (asking AI for general information related to the problem’s domain for the human to use for their sub-task), gathering methods information (asking AI for specific information on methods that the human is employing to solve their sub-task), and refining human-generated content (professionals provide their own output and using AI to refine its presentation).

Finally, a professional may engage in what we call “Abdicated Co-Creation,” (Self-Automator) relying on AI to determine both what should be done and how it should be done. In our setting, we were struck that roughly a third of the participants (27%; Self-Automators) were willing to abdicate co-creation throughout all phases of the human-LLM problem-solving process. This was particularly striking given that we studied highly skilled, highly conscientious professionals who reported to us that they took the experiment seriously, and who had been told that those who performed well would have this performance noted in their annual performance review. It is important to note that abdicated co-creation is not inherently problematic. For routine tasks, time-constrained situations, or problems well within AI's jagged frontier, delegation may be appropriate and efficient. Risks such as deskilling need to be considered in the context of the nature of the work. Future research should examine the contextual factors that make abdication helpful or harmful.

While earlier frameworks emphasize whether AI replaces or augments human labor, our findings reveal that GenAI introduces a third, more fluid possibility, Fused Co-Creation (Cyborgs), that transcends

this binary by dynamically redistributing agency between human and machine (Raisch & Krakowski, 2021). These findings also extend classic views of problem-solving as a socially embedded and coordinated process (Hargadon & Bechky, 2006; Okhuysen & Bechky, 2009). Where prior research depicts coordination and sensemaking as occurring exclusively among human actors, our study shows how professionals now draw AI into these processes, forming hybrid arrangements of collaboration and control. Co-creation modes represent not just technical configurations but new sociotechnical forms of distributed coordination, in which expertise, authority, and judgment are continuously negotiated between humans and AI. By theorizing these hybrid arrangements, we contribute to emerging research on human–AI collaboration by explaining how agency and expertise are jointly constructed in continuous interaction rather than statically allocated between human and machine. In this way, GenAI not only shifts how individuals approach problem-solving but also reshapes the collective architecture of expertise and coordination inside organizations.

In sum, we advance theories of human–AI collaboration by moving beyond recommendation-level interactions to conceptualize GenAI as a partner in continuous, conversational co-creation. Our findings demonstrate that GenAI reshapes the boundaries of collaboration, shifting it from discrete moments of input to an ongoing, dialogic process. Professionals can integrate GenAI throughout all phases of problem-solving, framing, analysis, iteration, and reporting, actively collaborating with it rather than receiving recommendations from it.

Human-AI Expertise Development

We find that each of these types has significant and different implications for professionals' expertise development around task-related and AI-related capabilities. Current literature on skilling details how individuals develop task-related expertise through formal education (e.g., Kaynak, 2019; 2024a; 2024b), rigorous on-the-job training (e.g., Beane, 2019, 2024; Beane & Anthony, 2024), occupational socialization (e.g., Van Maanen & Barley, 1984), and project-based work (e.g., Anthony, 2021). Yet the world of work is changing fast, and individuals need to rapidly develop skills. Recent research has shown that individuals may do this by senior professionals learning about new technologies

from junior professionals (e.g., Anthony, 2021; Beane, 2019, 2024; Beane & Anthony, 2024) or by attending “boot camps” (Kaynak, 2019; 2024a; 2024b).

We show that individuals can engage in rapid building of task-related expertise and AI-related expertise by interacting directly with AI. Yet, we also show that there is a tradeoff between building the two kinds of expertise. Professionals who engaged in Fused Co-Creation (Cyborgs) newskilled themselves in GenAI-related expertise and maintained their domain-related expertise. In contrast, professionals who engaged in Directed Co-Creation (Centaur) upskilled themselves in consulting domain-related expertise, but not in AI-related expertise. Those who engaged in Abdicated Co-Creation (Self-Automators) did not develop new expertise in either domain.

In short, co-creation directly shapes which skills professionals develop as they interact with generative AI: Fused Co-Creation (Cyborgs) develops AI-related fluency, Directed Co-Creation (Centaur) builds domain expertise, and Abdicated Co-Creation (Self-Automators) builds neither. These findings make clear that co-creation styles are not only about efficiency; they can be engines of skill transformation, determining whether professionals build AI fluency, deepen domain expertise, or risk losing both. Our findings thus position co-creation not merely as a form of collaboration but as a potential mechanism for longer-term skill transformation, reshaping how professionals learn, adapt, and sustain expertise.

Understanding Fused, Directed and Abdicated Co-Creation is critical to understanding the future of skilling in professional organizations. By reconceptualizing skilling as a skill-building process between humans and AI across a problem-solving workflow, rather than as a formal education process (e.g., Kaynak, 2019; 2024a; 2024b) or an on-the-job training process between individuals (e.g., Anthony, 2021; Beane, 2019, 2024; Beane & Anthony, 2024), this paper develops new theory about how and where professionals can develop skills.

New Facilitators of Human-AI Collaboration

Finally, we highlight new facilitators of human-AI collaboration. Prior research on human-AI collaboration has focused on barriers to rather than facilitators of human-AI collaboration. It highlights

three key barriers to human-AI collaboration: end-user professionals' experience of threat to autonomy, their frustration with AI opacity, and the difficulty of working with complex or low-usability systems (e.g., Barley, 1986; Mazmanian, 2013; Lebovitz, Lifshitz, & Levina, 2022). Current literature demonstrates both end-user rejection of predictive AI due to autonomy threats (e.g., Christin, 2017; Pachidi et al., 2021; Van den Broek et al., 2020, 2021; Valentine & Hinds, 2022) and frustration with predictive AI related to opacity (e.g., Anthony, 2021; Karunakaran, 2024; Lebovitz, Lifshitz, & Levina, 2022; Waardenburg, Huysman, & Sergeeva, 2022). In addition, scholars have shown that AI systems can be difficult to incorporate into professional routines because of their complexity and evolving design, which can create usability challenges and resistance to adoption (Bailey & Barley, 2011; Kaplan et al., 2017; Pachidi et al., 2021).

In contrast, our analysis highlights facilitators of human-AI collaboration. It shows that, with GenAI, rather than experiencing feelings of threat, frustration, or resistance, professionals may be enthusiastic about collaborating with AI. Further, their perceptions about GenAI may shape their mode of human-AI collaboration. Professionals who engage in Fused Co-Creation (Cyborgs) may do so because they perceive that collaborating with GenAI allows them to both develop AI-related expertise and increase their productivity. Professionals who engage in Directed Co-Creation (Centaur's) may do so because they perceive that collaborating with GenAI allows them to maintain control and avoid skill degradation. And professionals who engage in Abdicated Co-Creation (Self-Automators) may do so because they perceive that collaborating with GenAI allows them to increase their productivity while also allowing them to step back, because they are confident in the machine's ability to deliver results.

Boundary Conditions and Future Research

This work has several limitations. First, this was a single-session experimental task; we cannot assess how co-creation modes evolve over time or with repeated AI use. Second, we used GPT-4 from April 2023; AI capabilities have advanced substantially, potentially changing the dynamics we observed. This was also a study of high-performing individuals, which could bias toward self-automation, and it did not capture those who might refuse to co-create with GenAI at all. Moreover,

professionals in the study were junior-level consultants, and use by senior consultants may be different given that new technology is often used more heavily by juniors (Kellogg et al., 2024).

Taking this into account, we expect that these three types of co-creation and their associated outcomes will be most important in two organizational and institutional contexts that should be kept in mind when drawing on the concepts from this study. First, we expect to see these dynamics for tasks outside the frontier of GenAI current capabilities. At the time of our study, the problem-solving task participants completed was outside the frontier, but in the future and with other forms of GenAI, it may be different tasks that are outside the frontier. Second, we expect to see these dynamics for users who are using GenAI without other assistive interfaces layered on top. Future research could explore these boundary conditions.

Future research should also examine how these different modes of human–AI co-creation evolve as technologies improve and organizational practices adapt. While our study highlights GenAI as a collaborator in problem-solving, an open question is whether these systems will also take on more managerial functions, structuring tasks, directing workflows, or coordinating across teams. In this sense, scholars might extend our framework to explore the implications of algorithmic management in knowledge work (Koehler & Sauermann, 2024), investigating not only how individuals choose to work with AI but how AI itself may increasingly shape the organization of work.

Self-Automators present another avenue: our study explored professionals’ use of GenAI as an interactive partner, but future research should examine agentic AI, which fully automates parts of workflows. With agentic systems, the opportunity for interaction diminishes, raising questions about how co-creation and the very nature of work will change when professionals no longer remain “in the loop.”

Importantly, organizational goals also shape these choices. Firms that pursue human-centered goals, such as workforce development or preserving human involvement, tend to favor augmentation over automation (Koehler & Sauermann, 2024). This underscores that organizations are not only choosing technologies but also shaping the skill trajectories of their workforce: whether they risk hollowing out expertise through Abdicated use or reinforce human capabilities through Directed and Fused Co-Creation.

Practical Implications

Our findings have several practical implications for GenAI users and organizational leaders. For GenAI users, first, we show that users should engage with GenAI in different ways depending on their desired development of new expertise. In particular, professionals may want to engage in Fused Co-Creation if they are most interested in learning new GenAI-related capabilities. They may want to engage in Directed Co-Creation (Centaur) if they are most interested in learning task-specific capabilities. They may want to avoid Abdicated Co-Creation (Self-Automator) if they are interested in developing new GenAI-related or domain-specific skills. Whether professionals lean into AI as collaborators, treat it as a selective tool, or offload their judgment to it altogether will shape not only the outcomes of today's projects but also the very contours of expertise, authority, and competitive advantage within their organizations.

Second, even as we refer to participants as Cyborgs, Centaurs, and Self-Automators, we do not see Fused, Directed, and Abdicated Co-Creation as specific to specific individuals. Instead, the same individual may act as a Cyborg at some times and as a Centaur at others. We think that it is likely that individuals may have preferences for the use of particular styles, and this would be a great area for future research. Indeed, a key message of our paper is that individuals should try to step out of their preferred styles and adjust to the demands of the situation at hand. We hope that, by elaborating these different styles of co-creation and their associated outcomes, we can alert individuals both to their ability to choose to use various styles and to the imperative that they make these explicit choices rather than falling back on a style that is most comfortable to them.

For organizational leaders, we caution against using only the final outcome of human-AI collaboration (e.g., human edit rates or acceptance ratios) as a proxy for monitoring degree of user engagement versus automation complacency with GenAI. As we have shown, professionals who engage actively with GenAI through a Fused Co-Creation or Directed Co-Creation process may choose, in the end, to accept GenAI outputs or recommendations. Thus, to monitor potential automation complacency,

leaders may want to measure degree of interaction with GenAI rather than degree of editing or acceptance of AI outputs.

In sum, by elaborating three kinds of human-AI co-creation processes, Fused, Directed, and Abdicated co-creation, between the professional and AI throughout the full problem-solving workflow rather than focusing on the human-AI collaboration process at the point for recommendation, the present paper develops new theory on how humans can interact with AI and with what consequences for skill-building and the future of work.

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APPENDIX A

Appendix A Table 1: Business Analytics Subtasks

Subtask	Description
Problem Framing	Defining the scope of the case by identifying key issues, decision criteria, and questions that guide subsequent analysis.
Qualitative Analysis	Synthesizing insights from interview transcripts and textual data to surface themes, perspectives, and contextual factors.
Quantitative Analysis	Conducting calculations or interpreting numerical data (e.g., revenue tables) to establish patterns, trends, and financial implications.
Recommendation	Drawing together qualitative and quantitative insights to select among strategic options and propose a course of action.
Iterating	Revisiting earlier subtasks to refine outputs, test alternative approaches, or strengthen emerging arguments.
Extending	Generating additional ideas, strategies, or scenarios that go beyond the immediate analysis to widen solution space.
Evaluating	Assessing the robustness, feasibility, and risks of potential solutions to ensure they align with task objectives.
Reporting	Communicating the final analysis and recommendation in a clear, persuasive memo tailored to an executive audience.

Appendix A Table 2: Human-GenAI Interaction Practices

Mode of Human-GenAI Co-Creation	Practice	Example Prompts
Abdicated Co-Creation	Assigning a persona (Persona): Instructing GenAI to simulate a specific type of personality or character	● Rewrite as a recommendation to the CEO.
Abdicated Co-Creation	Requesting editorial changes to AI output (Editorialization): Asking GenAI to make editorial changes to the outputs AI has produced	● I like number 2 and a combination of 4 and 7. Can you rewrite this shorter list? ● This text is too long, please reduce it to 400 words. Be concise and only use the most important data points.
Directed Co-Creation	Mapping Problem Domain (Mapping): Asking AI for general information related to the problem's domain for the human to use for their sub-task.	● What are examples of clothing retailers that turned around a men's clothing brand after poor performance? ● Can you provide an example of a menswear clothing retailer that improved its in-store experience to drive revenue growth? ● What are the three factors to consider when evaluating which brand will drive revenue growth within a retail company.
Directed Co-Creation	Directing/Providing Methods tools information (Gathering): Asking AI for specific information on methods that the human is employing to solve their sub-task	● What is the Excel formula for CAGR? ● Write 50 words on how to define and target an appropriate customer segment.
Directed Co-Creation	Revising Human-Generated-Content (Revising): Professionals provide their own output and using AI to refine its presentation.	● Will you please review this for ease of reading, conciseness, and overall quality? The CEO should focus on re-vitalizing the men's brand at Kleding...
Fused Co-Creation	Teaching through examples (Teaching): Giving example of correct answer before asking AI a question	● Here is an example of the format I would like [paste paragraph], write in the same way.

Fused Co-Creation	Modularizing tasks (Modularizing): Breaking down tasks into multiple sub-steps for AI to execute	- First I am going to give you interview data to summarize. After I give you interview data, ask for the quantitative data. Once I give you the quantitative data, analyze it. After you analyze it, then provide a recommendation, to which I will then respond. After I respond, write a 500 word memo. - I will paste you the interview transcripts, for your context. Please do not start writing yet! I will give you more information, and then ask you for an updated version of the note to the CEO.
Fused Co-Creation	Validating: Asking AI to check its inputs, analysis, and outputs	- Can you make sure to revisit notes and make sure you added all important takeaways. - Please review your work. - Kleding's Financial Data: Kleding Man, Kleding Woman, Kleding Kids: Fiscal Year Total Revenues (in \$M), Total Revenues (in \$M), Total Revenues (in \$M), 2013 \$84.10 \$56.10 \$12.11...Does this data make sense to you?
Fused Co-Creation	Demanding logic explanation (Explanation): Asking AI to explain a confusing output; or why a particular recommendation was made	- Explain where does the 52.1% and 8% come from. - Can you further explain point 3 for me please?
Fused Co-Creation	Exposing Contradictions (Exposing): Pointing out logical or factual inconsistencies	Have you consider the fact that our men's brand is positioned as reasonable quality, fair price instead of trendy. Also, men in this age range is less digital savvy. Can you rewrite your recommendations please?
Fused Co-Creation	Elaborating: Asking AI to bring more breadth of details and nuance on an interesting or unexpected point	Can you elaborate on the rationale behind your recommendation?
Fused Co-Creation	Directing a Deep dive (Focus): Directing AI to focus on a particular data point, content or task	- Focus on Kleding Woman and Kleding Kids only. If you need to mention Kleding Man, use the reverse. - Focus the list on Kleding Man specifically.

Fused Co-Creation	Adding user's own data (Adding Data): Adding data after an output is generated to re-do the analysis in iterative cycles	• Please enhance the response above with the following new information from an interview:...
Fused Co-Creation	Pushing back: Disagreeing with the output and ask AI to reconsider	• I don't quite agree with your analysis, if we consider our strategy based on the highest impact per dollar /effort invested, wouldn't it be more impactful to bring the men's brand back to its original market share? You should also consider the fact that there is a high correlation between men's brand and child's brand. Please rewrite your suggestion • I don't quite agree with what you say here - I don't think most man would buy products from the female's brand and vice versa. This doesn't fit the culture of the market that we are focusing on.

Appendix A Table 3: Fused Co-Creation Newskilling Examples

Interaction	Professional's Prompt	GenAI Response	Fused Co-Creation Practice
1a	"Review your answer considering the following brand interviews...[provides interview transcripts]" Mateo Vargas	"Subject: Improving the Kleding Woman Brand to Drive Revenue Growth...[memo]"	Adding Data
1b	"Does your answer change considering what percentage of the each market Kleding already has?" Mateo Vargas	"Subject: Revised Recommendation - Focusing on Kleding Man for Revenue Growth...[memo]"	Exposing
1c	"Does a decrease in Kleding Kids change your answer?" Mateo Vargas	"Subject: Final Recommendation - Focusing on Both Kleding Man & Kleding Woman for Revenue Growth"	Exposing
Interaction 1 Newskilling: Mateo Vargas learned how to iteratively refine GenAI outputs by layering in additional context and data, first testing the system's initial recommendation, then questioning it with market share details, and finally probing with scenario changes, developing the skill to challenge, adjust, and co-create more robust analyses through targeted questioning.			
2a	"Can you please check yourself and redo the summarization. Aim for additional accuracy." Aisha Hassan	"Interview Summary - Daren F. Head of Product and Collections:...[summaries from interviews]"	Validating
2b	"Are you able to understand this data, on Kleding's Financial Data, as entered?" Aisha Hassan	"Yes, I can understand the financial data provided for Kleding's sub-brands."	Validating
2c	"Are you also able to understand this data, on Market Data, as entered?" Aisha Hassan	"Yes, I can understand the market data provided for the clothing market and its segments."	Validating
2d	"Using the two sets of data (Kleding's Financial Data and Market Data) entered above, can you provide 3-5 key insights about Kleding's finances as it relates to revenue growth across the three brands?" Aisha Hassan	"Kleding Man's revenue decline is particularly notable as the menswear market experienced significant growth over the same period. While the total menswear market grew from 45.20% in 2013 to 52.10% in 2017...[additional revenue analysis of Women's and Kid's brand]."	N/A
Interaction 2 Newskilling: Aisha Hassan learned how to validate GenAI's comprehension step by step, first by checking its summarizations, then confirming it could interpret structured financial and market data, ultimately developing the skill to probe system limits and ensure alignment between inputs, outputs, and the analytic task at hand.			

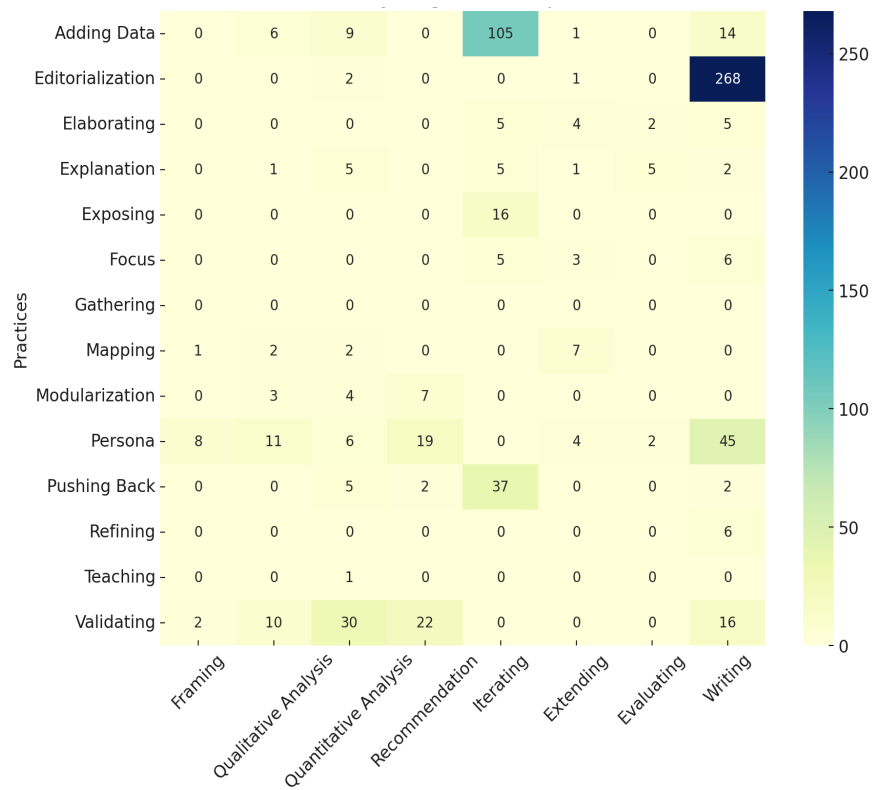
3a	“That makes some sense, but don't you think that the current momentum in Kleding Woman is valuable? Is there a risk that Kleding has lost their appeal with male customers due to their recent mistakes?” Lina Humphrey	You raise valid points. The momentum gained in Kleding Woman is indeed valuable, and it would not be advisable to overlook its importance while trying to address the challenges faced by Kleding Man. Kleding should continue nurturing the success of Kleding Woman to maintain growth and positive brand image within that target audience....”	Pushing Back
3b	“Provide counter arguments to this point.” Lina Humphrey	While the arguments provided are valid, there are potential counterarguments: 1. Diverting resources: Focusing on improving the Kleding Man product line might require diverting resources from Kleding Woman, which is currently performing well. This could potentially slow down the momentum Kleding Woman has built and negatively impact its success. were lost, and there is no guarantee of success in the process...[provides 4 more counterpoints.”	Explanation
3c	“Using this discussion that we've had, provide a PHD-economist level response to which product line (Kleding Man, Kleding Woman, or Kleding Kids) should the brand focus on going forward?” Lina Humphrey	As a PhD-level economist, I would argue that the optimal strategy for Kleding going forward is to adopt a multi-product differentiation approach that leverages the strengths and addresses the weaknesses of each product line. This implies that identifying a unique value proposition and targeting the most promising customer segments for each of Kleding Man, Kleding Woman, and Kleding Kids should be a priority. A nuanced and balanced focus on each product line can achieve long-term growth and stability, rather than relying on a singular strategy....”	Persona
3d	“While that makes a lot of sense, it feels like you haven't chosen a particular brand line to focus on. Let me present my hypothesis, and please provide some basic counter-arguments: The Kleding man line should be the focus for 3 reasons: 1. While Kledings recent sales have declined, there is likely a deep brand recognition that has been established	“Your hypothesis on focusing on the Kleding Man line is reasonable, and I understand your rationale. Here are some basic counter-arguments to consider: 1. Brand recognition vs. current perception...”	Pushing Back

	among men in the previous ~40 years of operations...” Lina Humphrey		
3e	“Please create a ~300 word summary of my hypothesis and the key points from your counter argument above.” Lina Humphrey	“Retail store layout is a critical factor that can influence customer behavior and sales. By strategically placing children's clothing near menswear, you can encourage men who are shopping for their children to browse and purchase items from the menswear section as well. Here are some store layout strategies to boost menswear sales:...”	Refining
Interaction 3 Newskilling: Lina Humphrey learned how to use GenAI as a sparring partner to sharpen reasoning, stress-test hypotheses, and refine arguments, by pushing back, soliciting counterarguments, and layering in expert personas, developing the skill to strengthen domain logic and craft more rigorous, persuasive recommendations through dialogic engagement.			

Appendix A Table 4: Directed Co-Creation Upskilling Examples

Directed Co-Creation Practice	Illustrative Data	Description
Gathering	“How to increase revenue on the female segment of an apparel store” Hana Al-Mansour	The professional upskills by asking GenAI for concrete methods on how to increase revenue, gaining step-by-step approaches they can apply in their analysis.
Gathering	“How to drive revenue sales growth for women's clothing.” Kasper Holm	The professional develops methodological expertise by using GenAI to suggest how to drive sales growth, learning targeted techniques to execute their task.
Gathering	“How can a company upskill staff with product knowledge in clothing stores?” Priya Deshmukh	The professional enhances problem-solving capability by asking GenAI how to upskill staff, gaining procedural guidance on training and development methods.
Mapping	“To rejuvenate a brand's image, what are best practices retailers can use?” Erik van Assendelft	The professional develops domain expertise by using GenAI to map out best practices in retail branding, expanding their knowledge of industry strategies.
Mapping	“What factors would you consider when choosing the ideal customer segment to prioritize in an apparel business?” Nadia Farouk	The professional develops domain expertise by asking GenAI for factors in customer prioritization, broadening their understanding of strategic decision-making in apparel.
Mapping	“What are examples of clothing retailers that turned around a men's clothing brand after poor performance?” Samuel Boateng	The professional develops domain expertise by prompting GenAI for turnaround case examples, gaining exposure to strategic lessons from industry precedents.
Mapping	“How many product types for ideal clothing store” Jonas Adler	The professional develops domain expertise by exploring product assortment considerations, deepening their understanding of how store configuration affects performance.

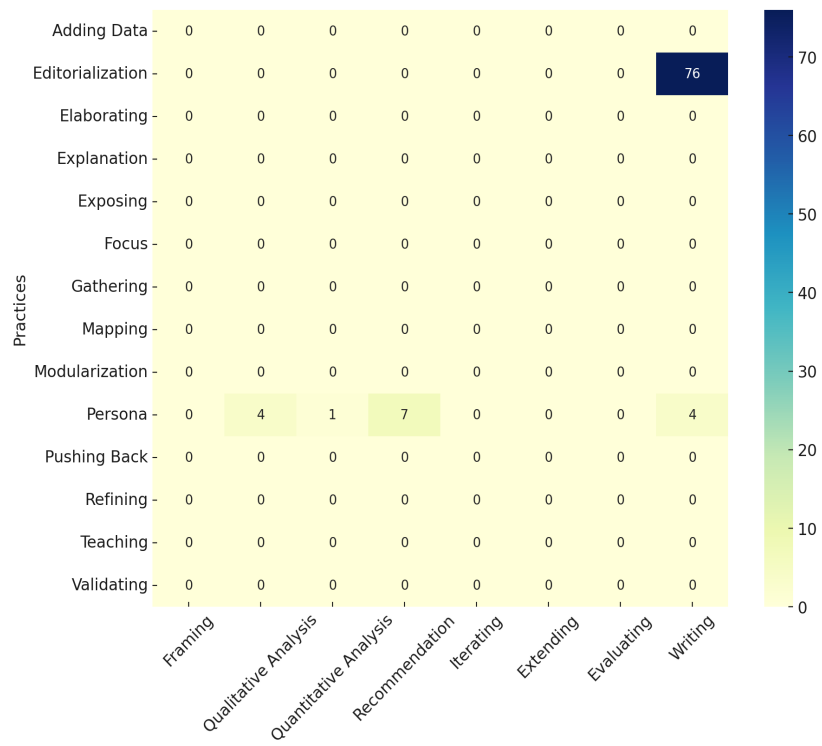
Appendix A Figure 1: Fused (Cyborg) Co-Creation Practice Map



Appendix A Figure 2: Directed (Centaur) Co-Creation Practice Map



Appendix A Figure 3: Abdicated (Self-Automator) Co-Creation Practice Map



APPENDIX B: Detailed Methods

Research Setting

This study was conducted in collaboration with a leading global management consulting firm as part of a larger research program on professional learning and technology adoption. The study was designed as a field experiment to examine how consultants use generative AI (GenAI) for problem-solving in realistic work conditions. The participants, 244 junior professionals (associates and consultants), were tasked with solving a strategic business problem using a custom-built “ChatGPT-like” platform that interfaced with OpenAI’s GPT-4 API (April 2023).

The platform captured all prompt inputs, AI responses, and timestamps for every session, creating a complete trace of human–AI collaboration.

Participants were asked to analyze data from a fictional retail company, drawing on interview transcripts with (fictitious) executives and historical performance data. They were instructed to recommend which of three brands (men’s, women’s, or kids’) should receive strategic investment and to present their recommendation in a CEO memo of no more than 500 words. Submissions were evaluated for correctness (binary: 1 = correct, 0 = incorrect; men’s brand was correct) and persuasiveness (1–10 scale, 10 = most persuasive).

Data Collection

Activity logs. We collected 244 complete conversational logs encompassing 4,975 human–AI interactions. Each log documented how participants engaged with GPT-4 across the workflow, providing granular insight into the micro-dynamics of human–AI collaboration. Logs captured every prompt and response, enabling temporal and functional mapping of the problem-solving process.

During post-task interviews, these logs were used as elicitation artifacts as interviewers referenced specific prompts or responses to probe participants’ reasoning, decision-making, and delegation patterns. We also linked logs to performance evaluations for correctness and persuasiveness.

Interviews. A team of 10 researchers with expertise in qualitative methods conducted 237 semi-structured interviews (Spradley, 1979), each approximately 60 minutes and held via Zoom within days of task completion. Interviews sought to uncover how professionals used GenAI and how they experienced this collaboration.

To elicit rich reflection, participants were shown artifacts from their sessions such as their own prompts and AI outputs, and were asked to explain why they issued certain instructions (for example, “Why did you push back on the AI here?”). The team also developed visual subtask maps that represented phases of the problem-solving process (for example, framing, analysis, recommendation), asking interviewees to indicate which steps they would assign to “human,” “AI,” or “both.”

Data collection was iterative and adaptive. Across eight collective debriefs, the research team refined the protocol based on emerging themes (Glaser & Strauss, 1967; Strauss & Corbin, 1990). For example, reframing questions about autonomy around managerial expectations (for instance, “Would you

use GenAI differently if your manager encouraged it?”) generated more nuanced responses. Midway, we paused data collection to conduct a deep inductive thematic analysis, which revealed early patterns in GenAI-supported problem-solving. This analysis led to five new questions focused on reasoning and task structuring with AI. Interviews resumed and continued for three additional team convenings. All interviews were recorded with participant consent and transcribed verbatim.

Data Analysis

Our data analysis followed an inductive theory-building process (Glaser & Strauss, 1967; Strauss & Corbin, 1990; Golden-Biddle & Locke, 2007) that proceeded through five phases.

Phase 1: Identifying Subtasks. We reviewed all 4,975 interaction logs to map the structure of human–AI interaction across the problem-solving workflow. Each interaction was coded into seven subtasks (Appendix A, Table 1): framing, analyzing data (qualitative and quantitative), recommending, iterating, extending, evaluating, and reporting. Transitions between subtasks were identified by shifts in intent or topic. For example, the prompt “Summarize the interview” was coded as qualitative analysis, while “Draft a CEO memo” was coded as reporting.

Phase 2: Coding Practices. Using open coding (Charmaz, 2014), we inductively revealed 14 interaction practices such as validating, mapping, modularizing, editorializing, and pushing back (Appendix Table 2). These codes captured how professionals directed, corrected, or collaborated with GenAI within each subtask.

Phase 3: Mapping Practices and Deriving Co-Creation Types. We then mapped practices to subtasks (Langley, 1999) to examine patterns of co-occurrence and sequencing. Using cleaned log data (random ID, subtask, practice label), we generated frequency tables in GPT-4 to assess the prevalence of each practice. Comparative analysis revealed three recurring interaction patterns: Fused, Directed, and Abdicated Co-Creation, distinguished by the degree and distribution of human versus AI control across the problem-solving process. We subsequently analyzed performance outcomes, accuracy, and persuasiveness by co-creation type.

Phase 4: Integrating Interviews. We analyzed interview transcripts inductively to link professional’s perceptions with the co-creation modes identified from the logs. This triangulation helped explain why professionals adopted specific collaboration styles and how they perceived the human–AI boundary.

Phase 5: Linking to Skilling. Finally, we examined how each co-creation mode related to skill development trajectories. We identified three patterns:

- Directed Co-Creation (Centaur) was associated with domain-specific upskilling.
- Fused Co-Creation (Cyborg) was associated with AI-related newskilling.
- Abdicated Co-Creation (Self-Automators) was associated with no discernible skilling.

This final step connected behavioral interaction patterns to professional learning outcomes, revealing how styles of GenAI collaboration shape not only performance but also expertise transformation.